Measurement of Inequality and Discrimination

Week 1: An Introduction to Causal Inference

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Course Details

- **Course:** The Measurement of Discrimination and Inequality
- **Office:** Ludwigstraße 31, Room 129 (office hours by appointment)
- **Seminar room:** Ludwigstraße 31, Room 021
- **Time:** Mondays, 16:00–18:00 c.t.

Requirements

1. 📚 Prepare for the seminars: do the readings and bring at least one question
2. 🗣️ Presentation (up to 15 minutes, 30% of grade) in week 12 (11 January); preferred topic by 30 November
3. 📝 Term paper based on the presentation (about 4000 words)

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Today

1. Why a course on discrimination and inequality starts with causation
2. Statistical data and linear regression
3. The problem of causal inference
4. The potential outcome model (Morgan & Winship, 2014, ch. 1–2)
5. Measuring discrimination
6. Readings for next week: causal graphs (Morgan & Winship, 2014, ch. 3–4)

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Discrimination and inequality of opportunity are causal

- Discrimination (on some definitions) occurs when a person's group membership causes her to be treated worse.
- Inequality of opportunity (on some definitions) occurs when factors outside of individuals' control cause unequal outcomes.
- Predictions by algorithms have been argued to be unfair **iff** some causal relation obtains between group membership and the prediction.

Does gender cause hiring decisions?

Suppose a firm hires a larger share of its male applicants than of its female applicants.

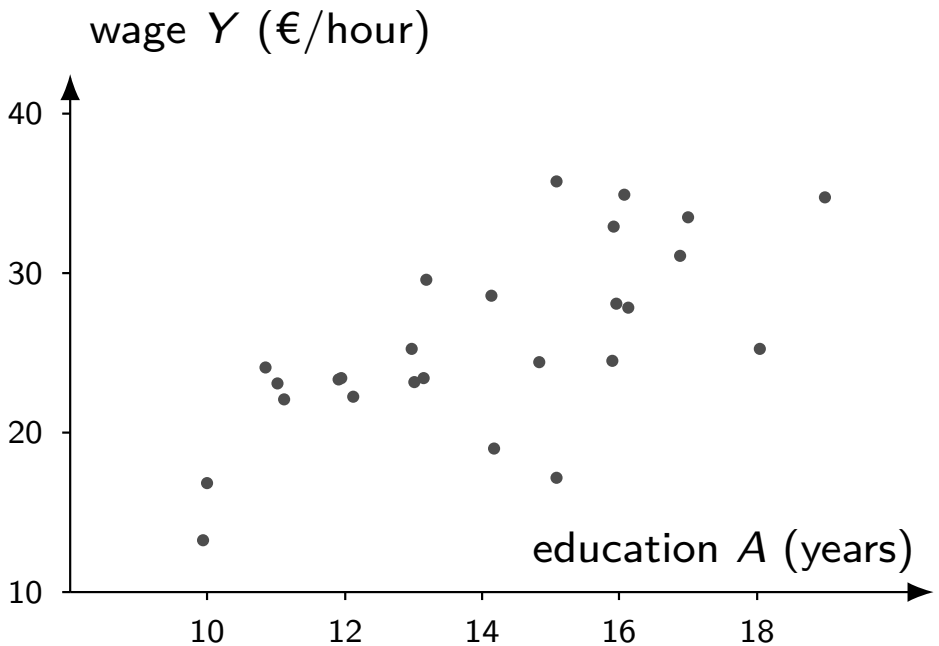
- Is that discrimination? 🤔
- The hiring numbers alone cannot tell different explanations apart.

Data: a table of individuals and variables

A typical social-science data set (made-up numbers):

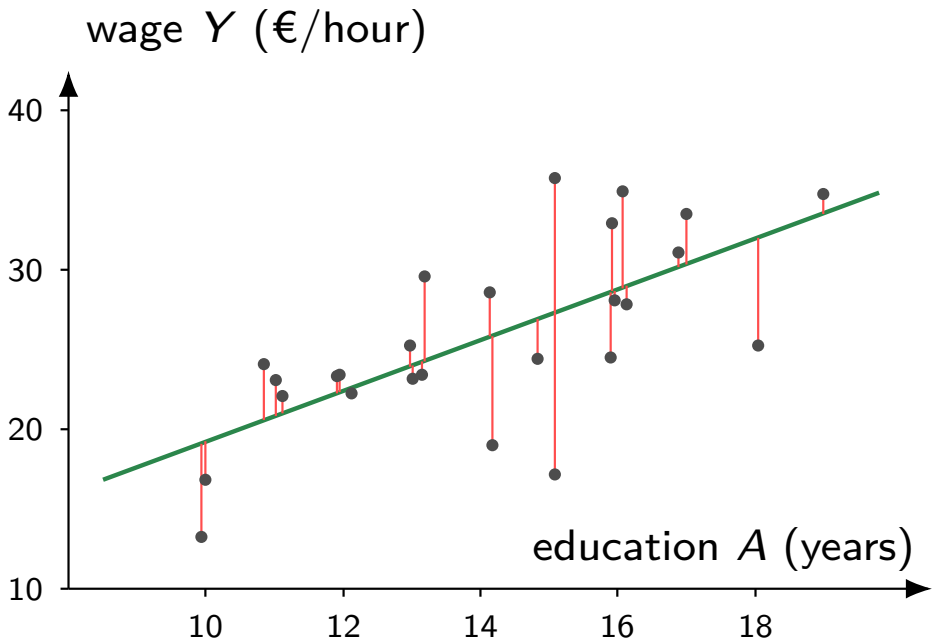
person i	wage Y_i (€/hour)	education A_i (years)	experience B_i (years)
1	32.90	16	12
2	34.90	16	20
3	27.80	16	5
4	31.10	17	9
5	22.10	11	15
⋮	⋮	⋮	⋮

Looking at two variables: a scatterplot



- Each dot is one person: how many years of education they had, and what they earn.
- People with more education tend to earn more: the variables are **correlated**.

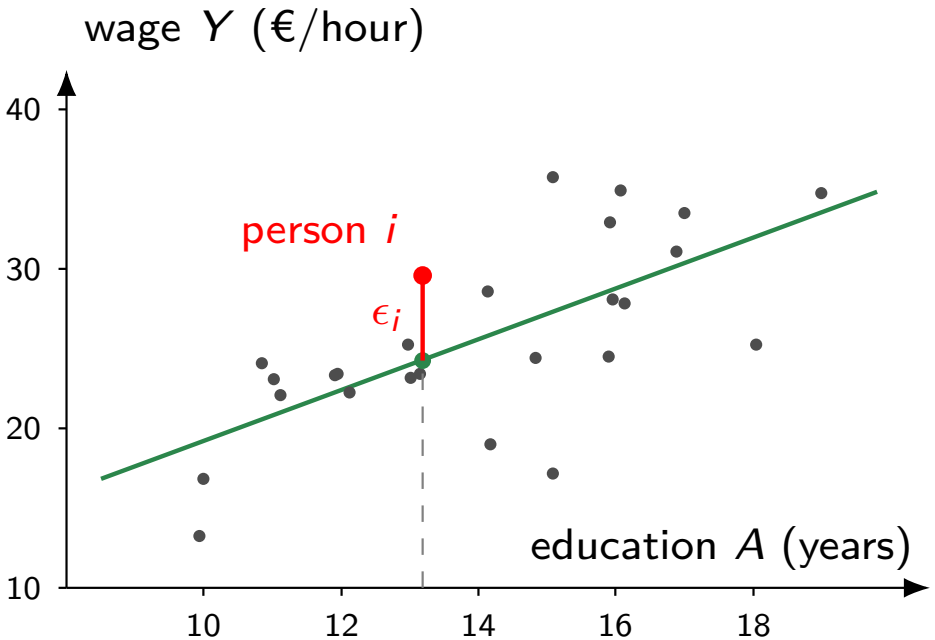
Linear regression: the line that fits best



- Draw a straight line through the cloud:
  predicted wage $= c + \alpha \cdot$ education
- For each person, the prediction is off by some amount: the **error** (red lines).
- **Linear regression** chooses the line (the numbers c and α) that makes the errors **as small as possible** overall.

The green line has $c = 3.3$ and $\alpha = 1.6$.

One person, one equation



For every person i:

$$Y_i = \underbrace{3.3 + 1.6A_i}_{\text{prediction}} + \underbrace{\epsilon_i}_{\text{error}}$$

Person i has

- $A_i = 13.2$ years of education
- predicted wage $3.3 + 1.6 \cdot 13.2 \approx 24.4$
- actual wage $Y_i = 29.5$
- so error $\epsilon_i = 29.5 - 24.4 = 5.1$

ϵ_i: how much more (or less) i earns than we would predict from education alone.

More than one variable

With two explanatory variables, education A and experience B:

$$Y_i = c + \alpha A_i + \beta B_i + \epsilon_i$$

- **A data point** $(Y_i, A_i, B_i, \epsilon_i)$ describes one person i:
  - Y_i, A_i, B_i: i's wage, education and experience. These are *observed*.
  - ϵ_i: the part of i's wage that education and experience do not predict.
- **Meaning of** α: among people with the *same* experience, one more year of education goes with α more euros per hour on average.

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One equation, two readings

$$Y_i = c + \alpha A_i + \beta B_i + \epsilon_i$$

Statistical reading

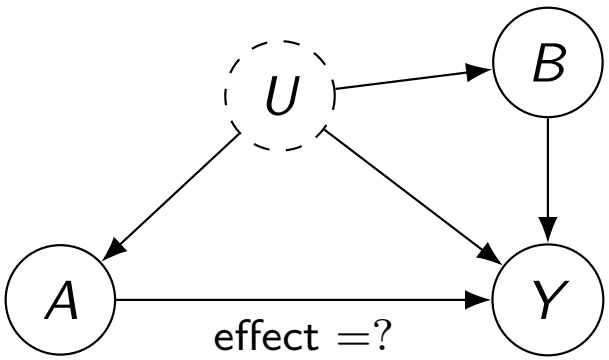
- describes the data
- α: how wage and education go together
- ϵ_i: the prediction error
- useful only for **prediction** in the same population

Causal reading

- describes how wages are *produced*
- α: the *effect* of one more year of education on wage
- ϵ_i: all *other causes* of i's wage
- tells us what *would* happen to i's wage if i's education were changed (e.g. by a policy)

⚠ **Regression almost always gives you the statistical α only!**

Why the two can come apart



A = education, Y = wage, B = experience, U = ability, family background, ...

! Unobserved factors U may cause the other variables. If U affects both A and Y, the statistical α is not the causal α. Names for this bias:

- Confounding
- Endogeneity
- Omitted variable bias

The identification problem

Statistics is not causal inference. (Ackermans, 2023)



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Causal questions are what-if questions

- **Causal question**: did Anna's degree raise her earnings?
- Suppose:

Anna ...	earnings
with a degree (actual)	€4,000
without a degree (counterfactual)	€3,000
effect of the degree on Anna	€1,000

This definition of an "effect" is the core idea of the potential outcome model (Morgan & Winship, 2014, §1.1).

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The potential outcome model

- A population of individuals i
- A cause with two **causal states**: **treatment** $(D = 1)$ and **control** $(D = 0)$
- Each individual has two **potential outcomes**:
  - y_i^1: i's outcome *if* i were treated (Anna: 4,000)
  - y_i^0: i's outcome *if* i were not treated (Anna: 3,000)
- The **individual causal effect** of the treatment on i is the difference:

$$\delta_i = y_i^1 - y_i^0 \qquad \text{(Anna: } 4{,}000 - 3{,}000 = 1{,}000\text{)}$$

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Defining the causal states

- **Fine articulation.** All states have clear definitions.
- **Reasonable counterfactuals.** The counterfactuals are not so complicated to have no clear truth value.

Question

❓ What are the causal states when we ask about the effect of *gender* or *race*? What does "y_i^0: i's wage if i were a man" mean?

A God's-eye view (hypothetical data)

Treatment D: college degree. Outcome: monthly earnings (in thousands of euros).

i	d_i	y_i^1	y_i^0	$\delta_i = y_i^1 - y_i^0$
1	1	12	7	5
2	1	10	6	4
3	1	8	5	3
4	0	9	6	3
5	0	7	5	2
6	0	6	4	2

What we actually see

For each person we only see the outcome under the state they are actually in:

i	d_i	y_i^1	y_i^0	δ_i	observed y_i
1	1	12	?	?	12
2	1	10	?	?	10
3	1	8	?	?	8
4	0	?	6	?	6
5	0	?	5	?	5
6	0	?	4	?	4

The fundamental problem of causal inference (Holland, 1986)

For each individual we observe at most one potential outcome. The other is *counterfactual*. Individual causal effects can never be observed directly.

The average treatment effect (ATE)

The **average treatment effect** is the average of the individual causal effects over everyone in the population:

$$\text{ATE} = \mathrm{E}[\delta] = \mathrm{E}[Y^1 - Y^0]$$

i	d_i	δ_i
1	1	5
2	1	4
3	1	3
4	0	3
5	0	2
6	0	2

In our example:

$$\text{ATE} = \frac{5+4+3+3+2+2}{6} \approx 3.17$$

On average, a degree raises monthly earnings by about €3,170.

But we cannot compute this from the data 😢

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The naive estimator (Morgan & Winship, 2014, §2.7)

- The NE: the average outcome of the treatment group minus the average outcome of the control group.

$$\underbrace{\frac{12+10+8}{3}}_{\text{graduates}} - \underbrace{\frac{6+5+4}{3}}_{\text{non-graduates}} = 10 - 5 = 5$$

- The NE is 5. The true ATE is 3.17.

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Why randomised experiments allow causal inference

ℹ️ In randomised controlled trials (RCTs) we decide who gets the treatment by flipping a coin. As a result:

1. The treatment group and the control group are, on average, alike in every causal factor other than the treatment: ability, background, motivation, ...
2. There are no "confounders", causes of both the treatment and effect.

(By *Reichenbach's Common Cause Principle:* 2 implies 1.)

👉 In this case (given some more assumptions), the NE approaches the ATE given enough data.

Without experiments

With observational data, we need **causal assumptions**. Main strategies:

1. **Back-door conditioning** (comparing like with like): find all common cause variables of the treatment and effect and compare groups that are alike on these variables.
2. **Natural experiments**: find a different cause of the outcome, via the treatment, that has no common causes with the outcome.

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An experiment: sending fictitious CVs

🔎 **A CV audit study of gender discrimination** (Birkelund et al., 2022):

- 2016–2018: 4,279 fictitious applications to real job ads in six countries
- jobs measured: cook, payroll clerk, receptionist, sales representative, software developer, store assistant
- Gendered names are assigned at random to CVs
- Outcome: does the employer reach out?

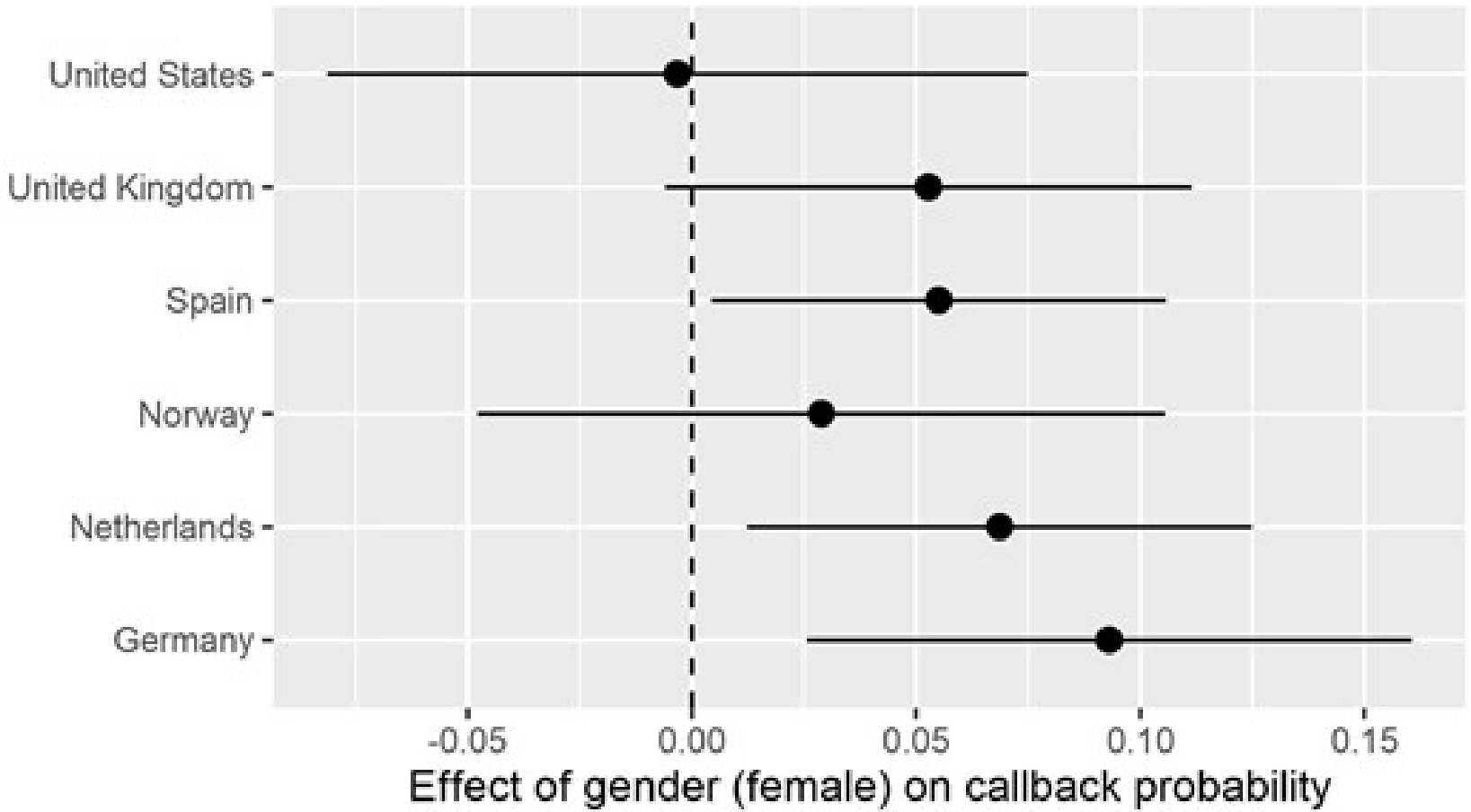
What the experiment found

- No discrimination against women in any of the six countries.
- Some discrimination against men, mostly in female-dominated jobs.

Germany: share of applicants called back	men	women
receptionist	57%	79%
store assistant	25%	48%

Birkelund et al. (2022, Table 3)

What the experiment found



Source: Figure 1 in Birkelund et al. (2022)

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Another experiment, another answer

🔎 **Another CV study in Spain** (González et al., 2019):

- 2016: Four CVs (two male, two female) sent to 1,372 job ads
- A very wide range of jobs

Results:

- Discrimination against women on average
- No discrimination against highly qualified women with no children

Share called back	men	women
all applicants	10.9%	7.7%
without children	11.5%	8.8%
with two children	10.3%	6.6%

González et al. (2019, Table 1)

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What exactly is the counterfactual?

Hu and Kohler-Hausmann (2020, 2025) criticise CV audit studies.

- The usual interpretation: CV studies measure the perception of gender (Greiner & Rubin, 2011).
- But how do we compute the counterfactual: "What if the employer had perceived a man?"
- Hu and Kohler-Hausmann: gender and other features of the applicant are often *co-constituted*.

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Discussion

1. Does "the effect of being a woman on wages" have well-defined potential outcomes?
2. Can a person be discriminated against even if the counterfactual comparison shows no difference?
3. Is a gender pay gap that is *fully explained* by occupation thereby *not* due to discrimination?

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Next week: causal graphs

Required reading: Morgan and Winship (2014), chapter 3 and sections 4.1–4.2.

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§3.1 Identification

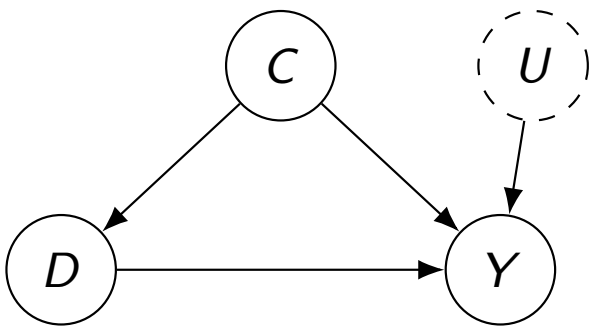
Two problems of causal inference (Morgan & Winship, 2014):

1. **Statistical inference**: how good are the estimated population parameters?
2. **Identification**: could we compute the causal effect given high-quality data about the population?



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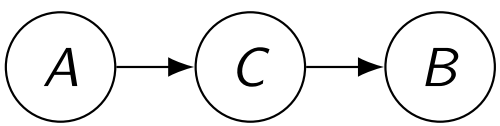
§3.2.1 The elements of a causal graph



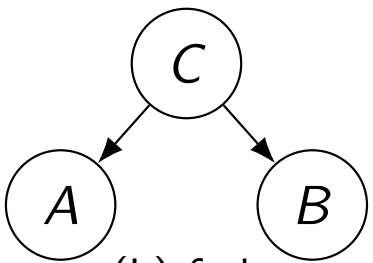
- **Nodes**: variables.
- **Arrows**: causal relations.
- **Paths**: any chain of arrows.
- **Directed paths**.
- **Parents/children**.
- **Ancestors/descendants**.

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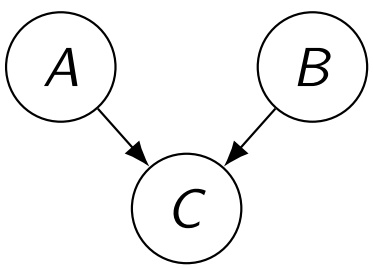
§3.2.2 Three basic patterns



(a) chain
(mediation)



(b) fork
(common cause)



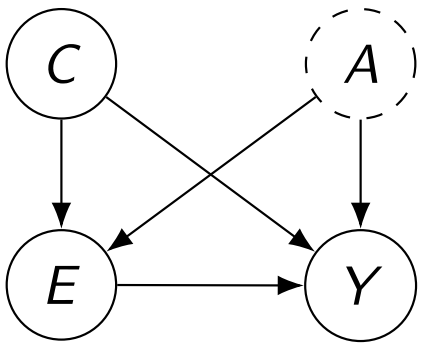
(c) collider
(common effect)

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§3.4 Seeing versus doing

- Pearl distinguishes two situations:
  - **seeing**: we *observe* who has $D = 1$
  - **doing**: we *set* $D = 1$, written $do(D = 1)$
- Causal effects are defined by *doing*: compare average Y under $do(D = 1)$ with average Y under $do(D = 0)$.
- This is the same as comparing Y^1 with Y^0 (the ATE).
- When there is confounding, *seeing* differs from *doing*.

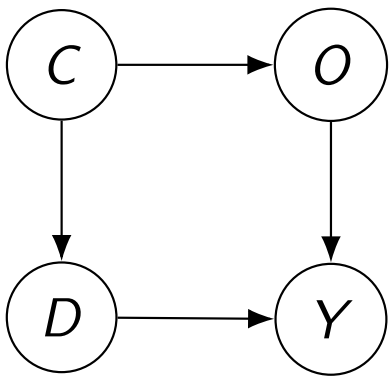
§3.2.3 Confounding and back-door conditioning



E = education, Y = earnings, C = family background, A = ability. Morgan and Winship (2014, fig. 3.5)

- C and A are **confounders**.
- **Back-door conditioning** on C: compare people with the same family background.
- This removes the confounding by C, but not A (unobserved).

§4.1.1 Back-door paths



Morgan and Winship (2014, fig. 4.1)

- A **back-door path** from D to Y is a path that starts with an arrow *into* D: $D \leftarrow C \rightarrow O \rightarrow Y$.
- Back-door conditioning: "block" all back-door paths.
- Conditioning on C, on O, or on both blocks the back-door path.

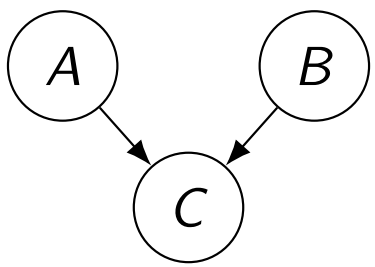
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§4.2 The back-door criterion

The back-door criterion: **block every back-door path, do not open new ones, and leave the causal path of interest alone.**

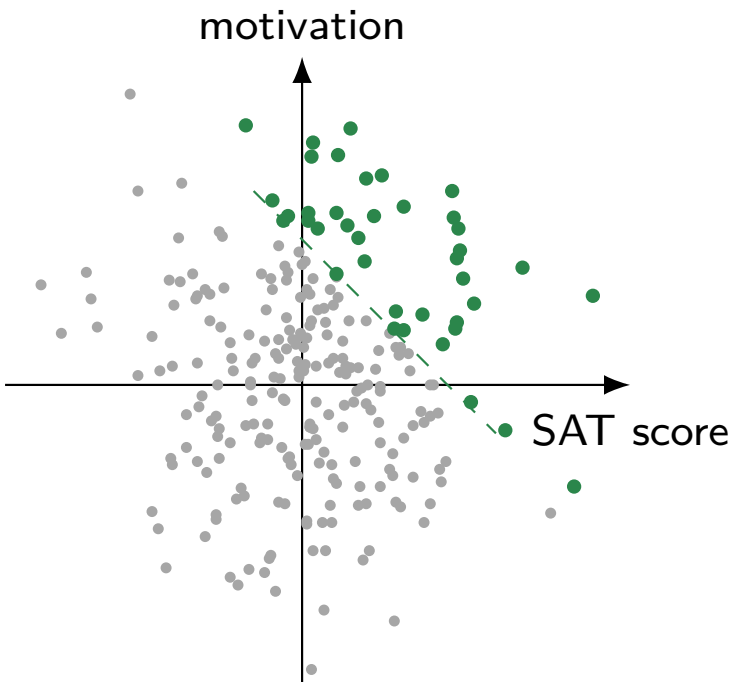
A path is **blocked** iff it contains

- a chain $\to C \to$ or a fork $\leftarrow C \to$ with C conditioned on, **or**
- a collider $\to C \leftarrow$ where neither C nor any descendant of C is conditioned on;



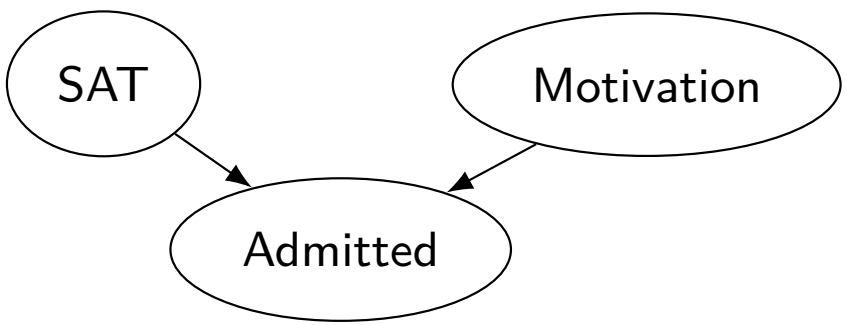
Collider

§4.1.2 Conditioning on a collider



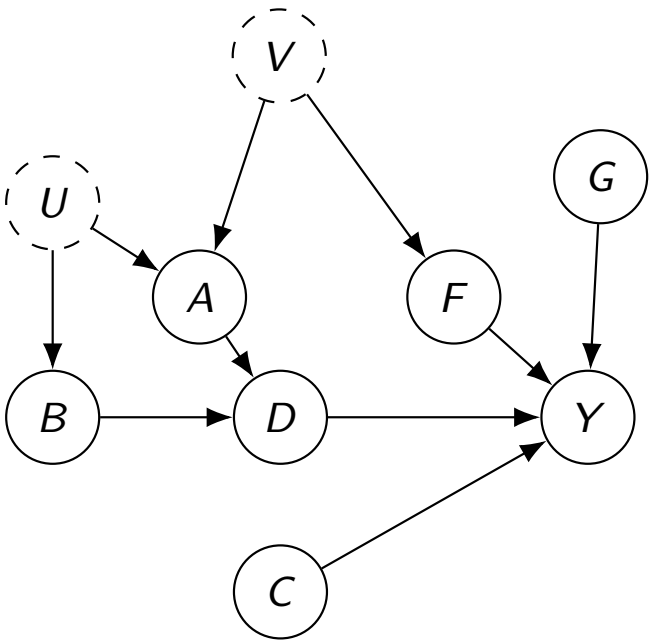
250 simulated applicants; green = admitted

A college admits applicants based on SAT score *and* motivation:



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Exercise



after Morgan and Winship (2014, fig. 4.4)

We want the effect of D on Y.

1. List the back-door paths from D to Y.
2. On which path is A a collider?
3. Does conditioning on F identify the effect?
4. Does conditioning on A alone?
5. What about A, B, and F together?

Answers are in §4.2 of the reading.

References I

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